



Video · youtu.be/hws8kKtEp44

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HOW IT WORKS

- 1 Reports become evidence.** Each report is snapped to a stream network built from OpenStreetMap, with flow direction, culverts and bridged gaps.
- 2 One belief over what and where.** An exact joint probability over six explanations and every candidate outfall. Weather (Open Meteo) sets the priors: dry days favour a misconnection, rain an overflow.
- 3 Clean results count.** A clean look lowers every source upstream of it. Discharges are modelled as intermittent, so a clean check never rules a source out.
- 4 The next best check.** Every check at every point is scored:

$$\text{EVSI} + 1.2 \cdot \text{bits} - \text{cost} - 0.1 \cdot \text{delay}$$

Citizen missions are also charged for the walk from where the person reported.
- 5 Photos help, people decide.** EXIF is stripped and faces are blurred before a vision model (Claude or Gemini) reads a photo, which counts as a weaker observer than any person.

ARCHITECTURE

One container, deployed on Render. **Engine:** Python, NumPy, NetworkX. **API:** FastAPI with roles, hashed tokens, rate limits, a strict CSP and an append only SQLite event store that rebuilds cases by replay. **Web:** React and MapLibre PWA. **CI:** 84 tests, accessibility audit, FHIR validation, image build and smoke test.

STATUS AND NEXT STEPS

Built and tested	Next, with a pilot partner
Engine, recommender, SourceBench and scenario replay	Real outfall inventories instead of synthetic candidates
Staff roles and approvals, audit trail, public advisories	Expert review of likelihoods; calibration against lab results
FHIR on the OneAquaHealth IG, validated and pushed	Overflow telemetry and sensor feeds as evidence
Accessible, offline capable citizen app in four languages	Native review of translations; notifications when a case changes

Data honesty. Stream geometry and places (OpenStreetMap) and weather (Open Meteo) are real. The demo's candidate outfalls, citizen reports and check results are simulated and labelled that way on screen. SourceBench results are from simulation, not field trials.

RESULTS · SOURCEBENCH (SIMULATION)

Strategy	Correct	Median checks	Cost
Walk the bank	28%	20	1.49
Bisect the stream	43%	13	0.95
Dipper	56%	9.5	1.30
Pure information gain	74%	8	4.92

7 stream networks (5 real OneAquaHealth streams, 2 synthetic), 40 trials each, 25-check budget, a simulated world noisier than the model. Dipper doubles the success of a bank walk at lower cost; pure information gain spends about 3.8 times more.

STANDARDS AND TRUST

- FHIR:** nine Dipper profiles on the OneAquaHealth IG (LocationOah, GroupOah), written in FHIR Shorthand, 0 validator errors in CI. Idempotent transaction push.
- People approve:** advisories, handovers and dismissals need a signed-in person with the right role; every decision is in an audit trail.
- Privacy:** anonymous citizens (keyed pseudonyms), private report tokens never in URLs, faces blurred, no IP addresses in logs.
- Accessibility:** 0 axe violations (WCAG 2.2 AA) at desktop, phone and 320 px.